

Amulet Motion Controller

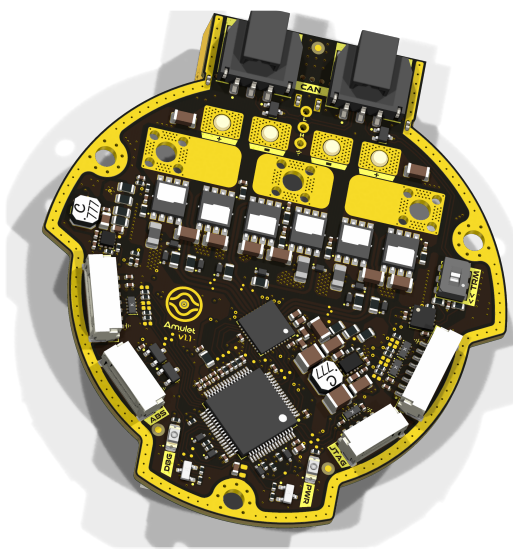
Variant:

2026-08-17

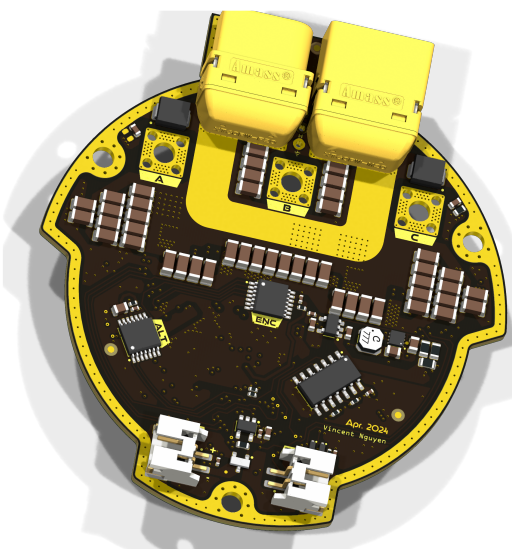
Rev 1.1.1+ (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 17-Aug-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for cautionary design notes.

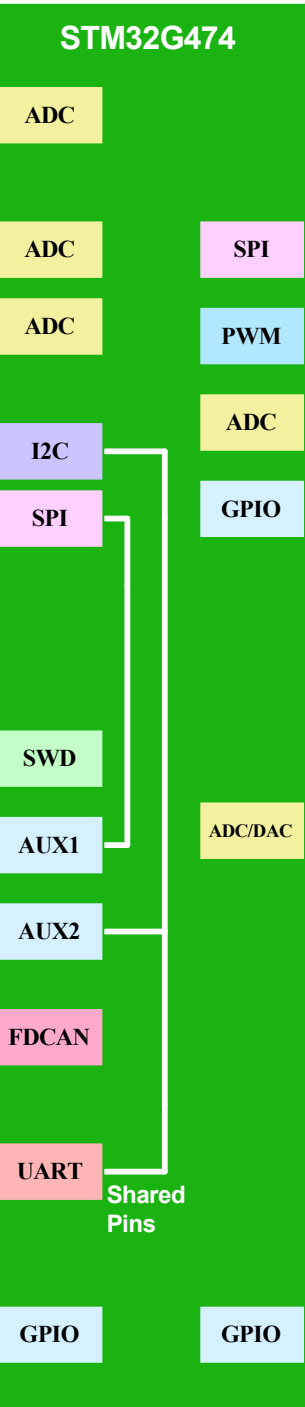
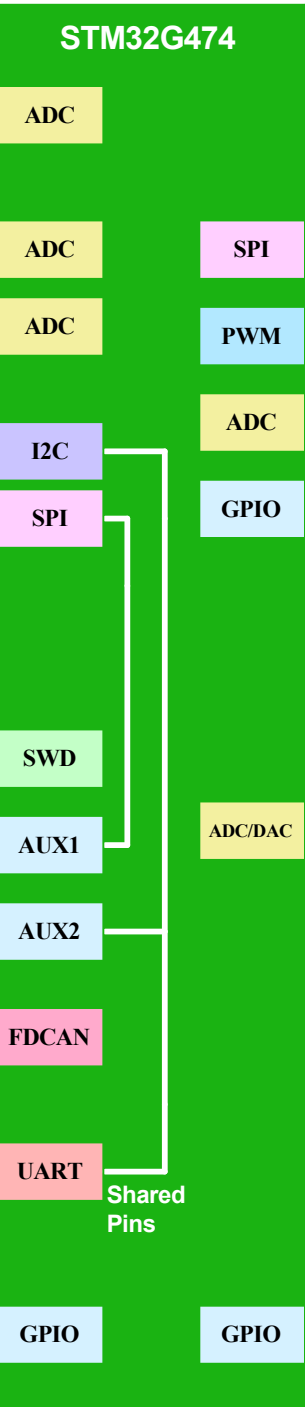
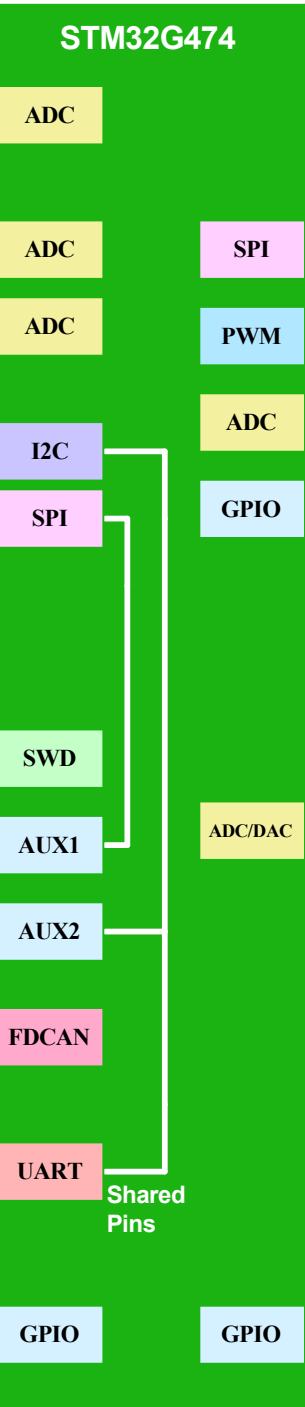
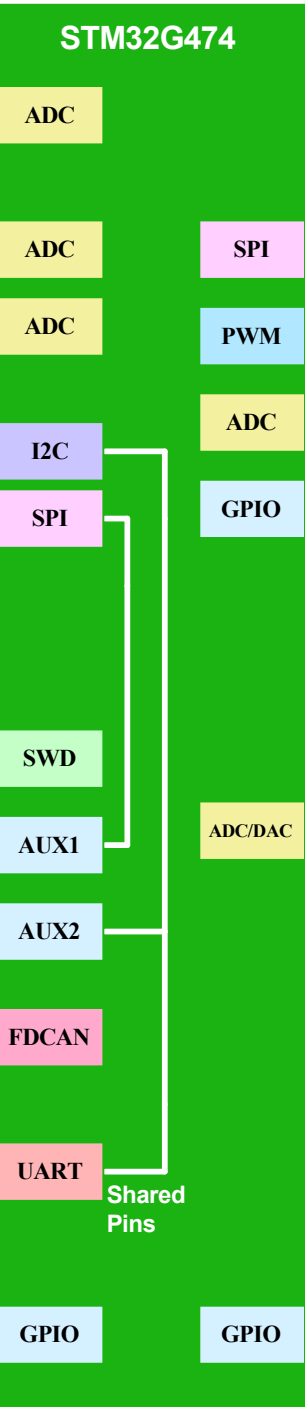
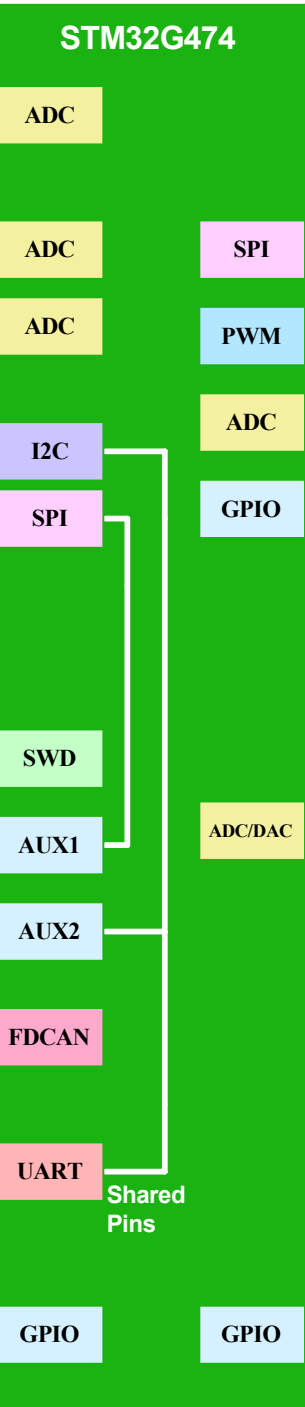
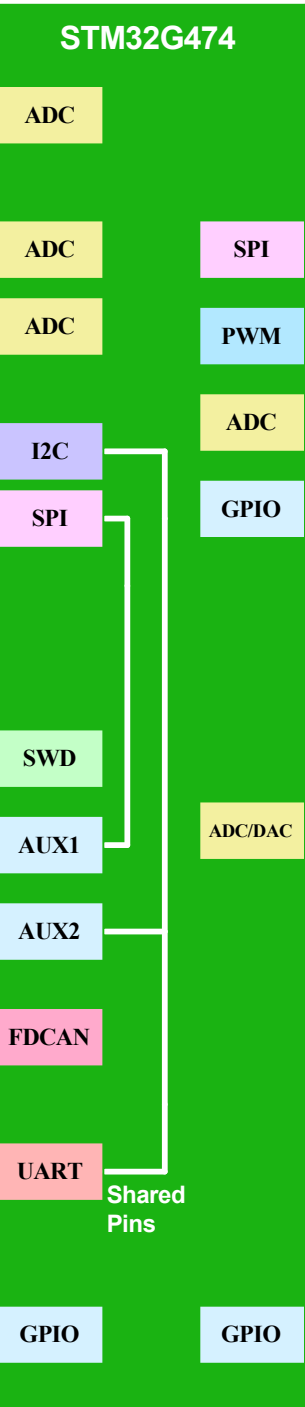
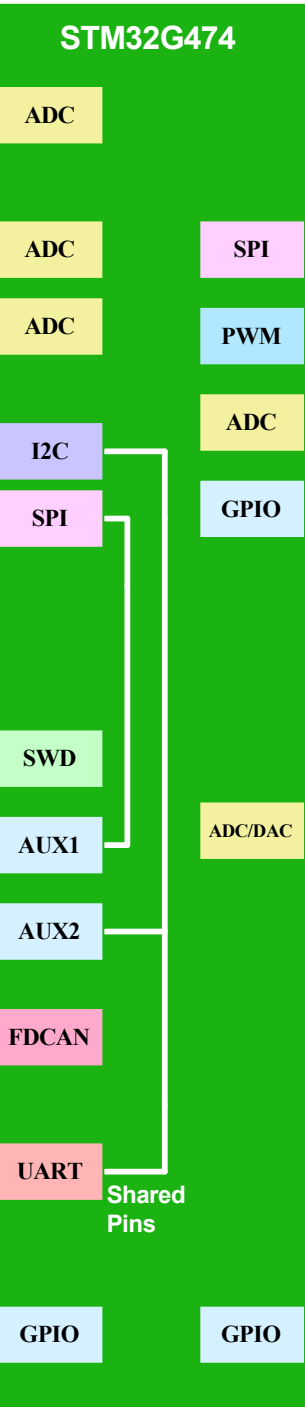
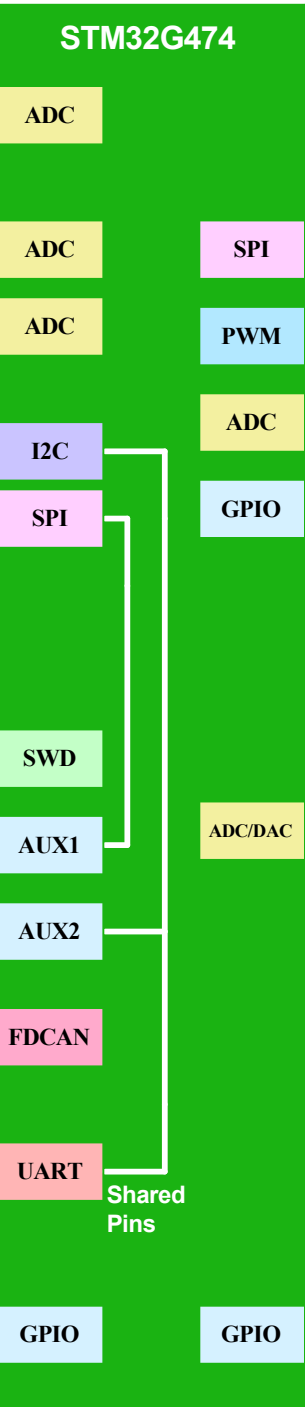
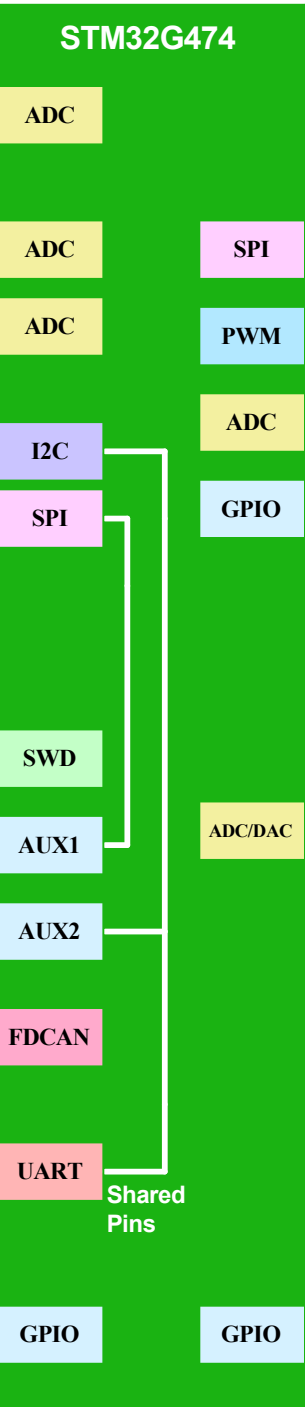
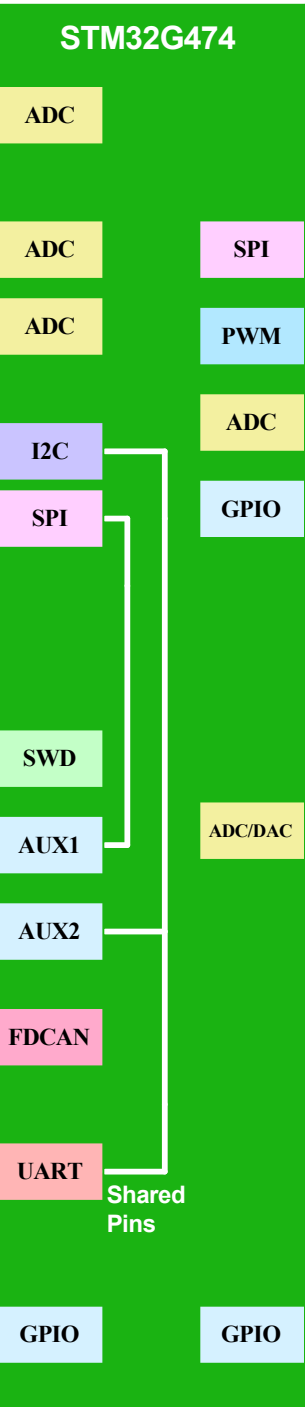
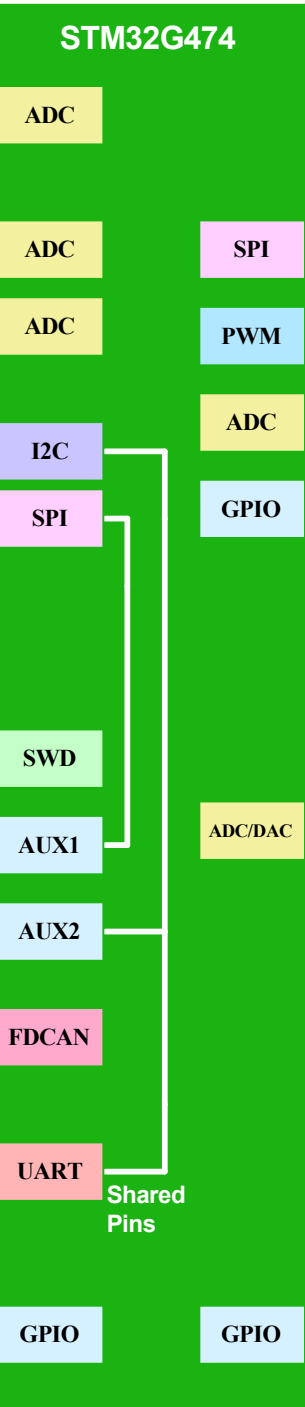
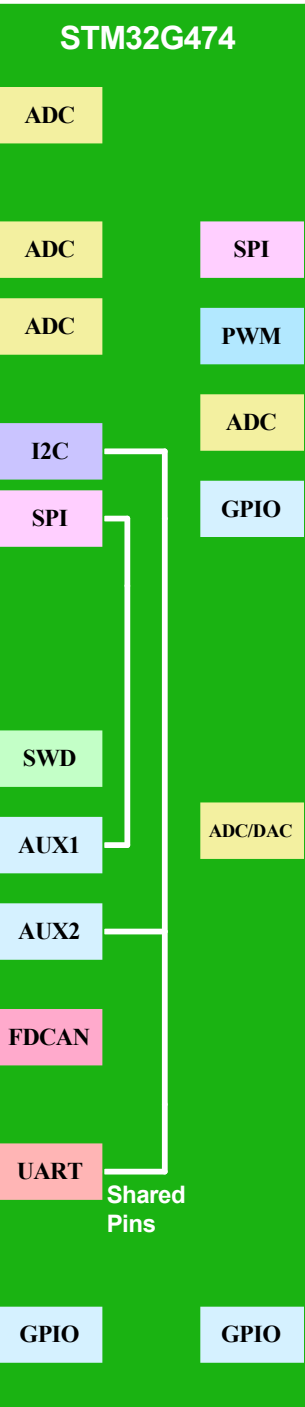
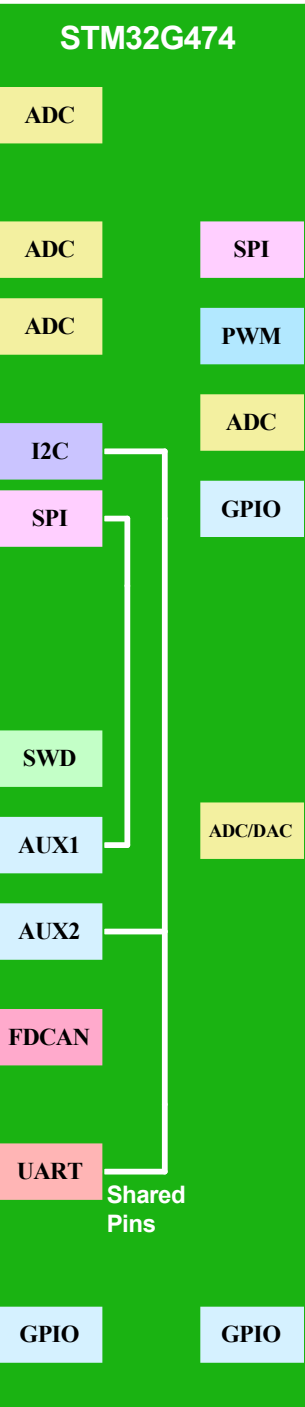
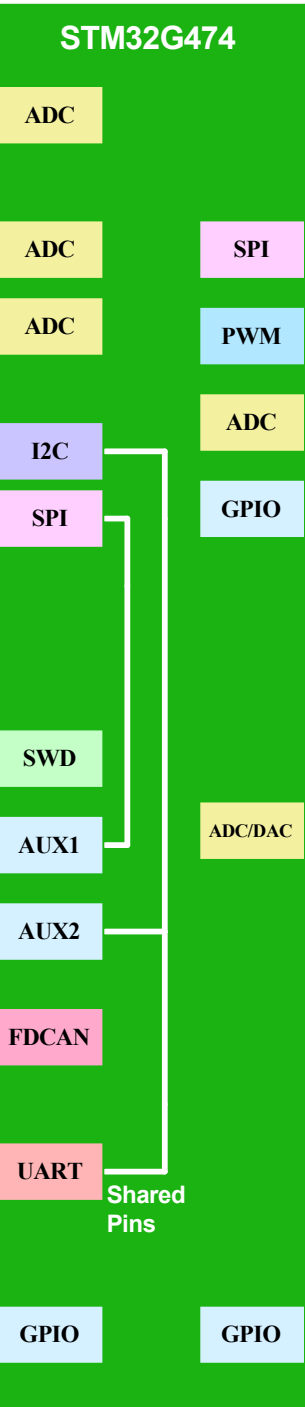
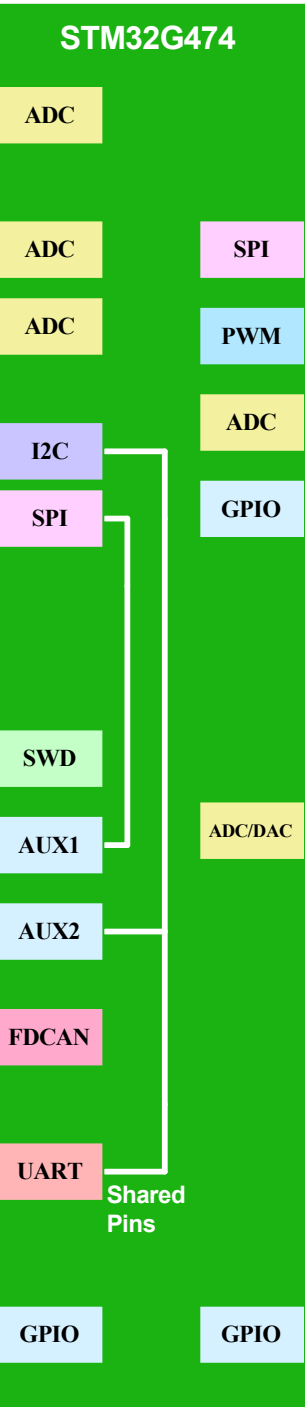
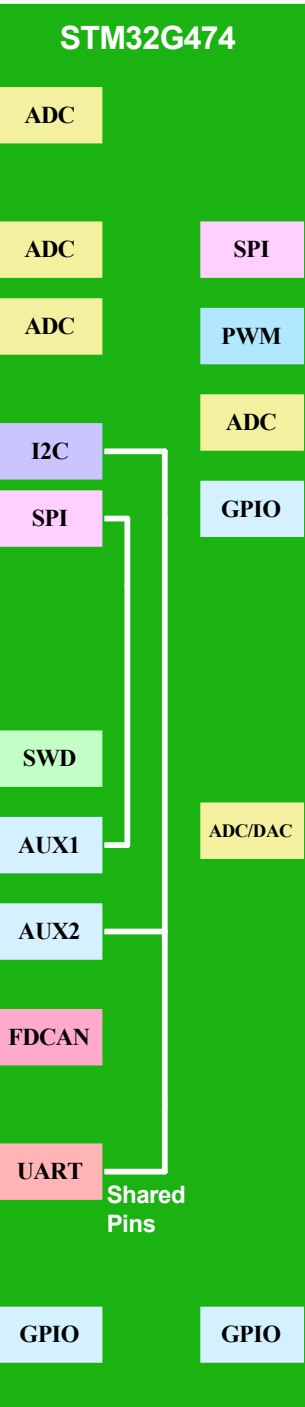
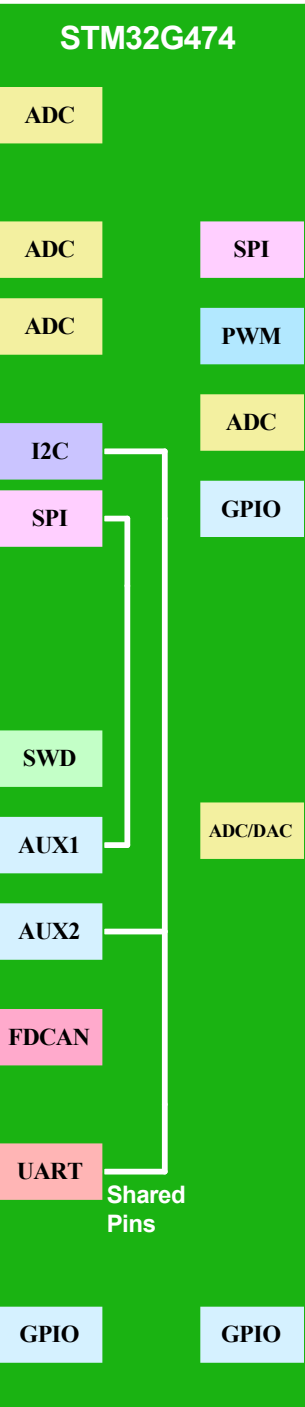
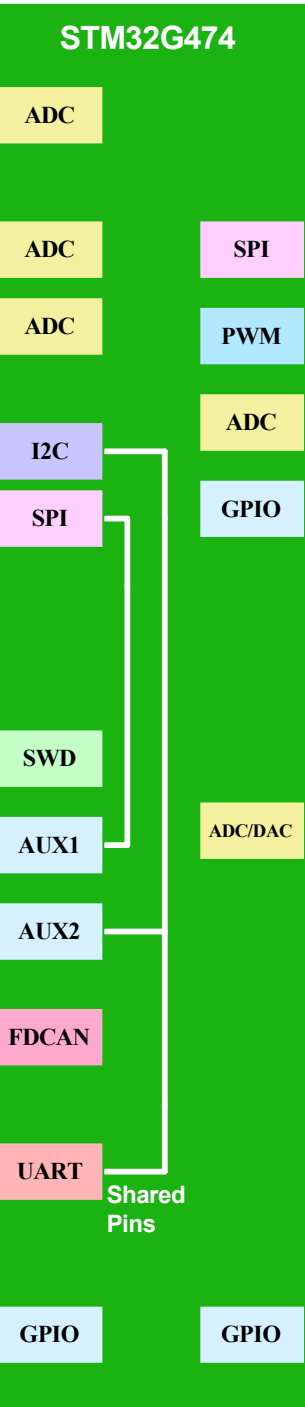
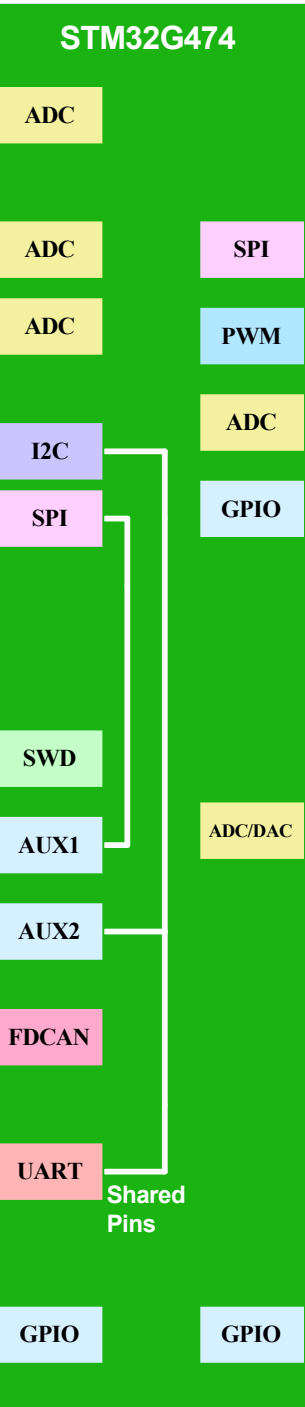
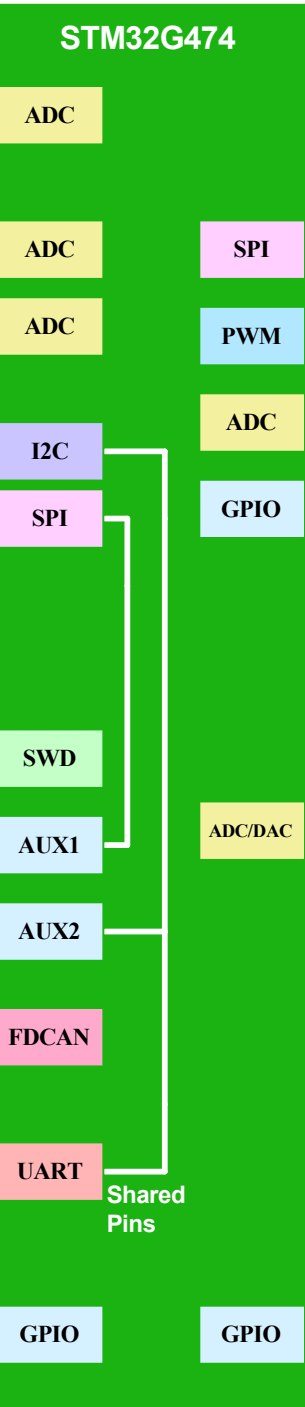
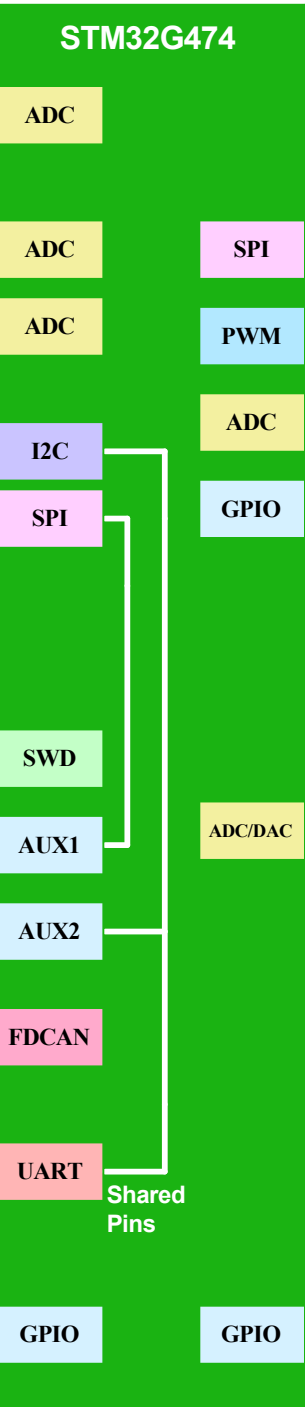
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Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

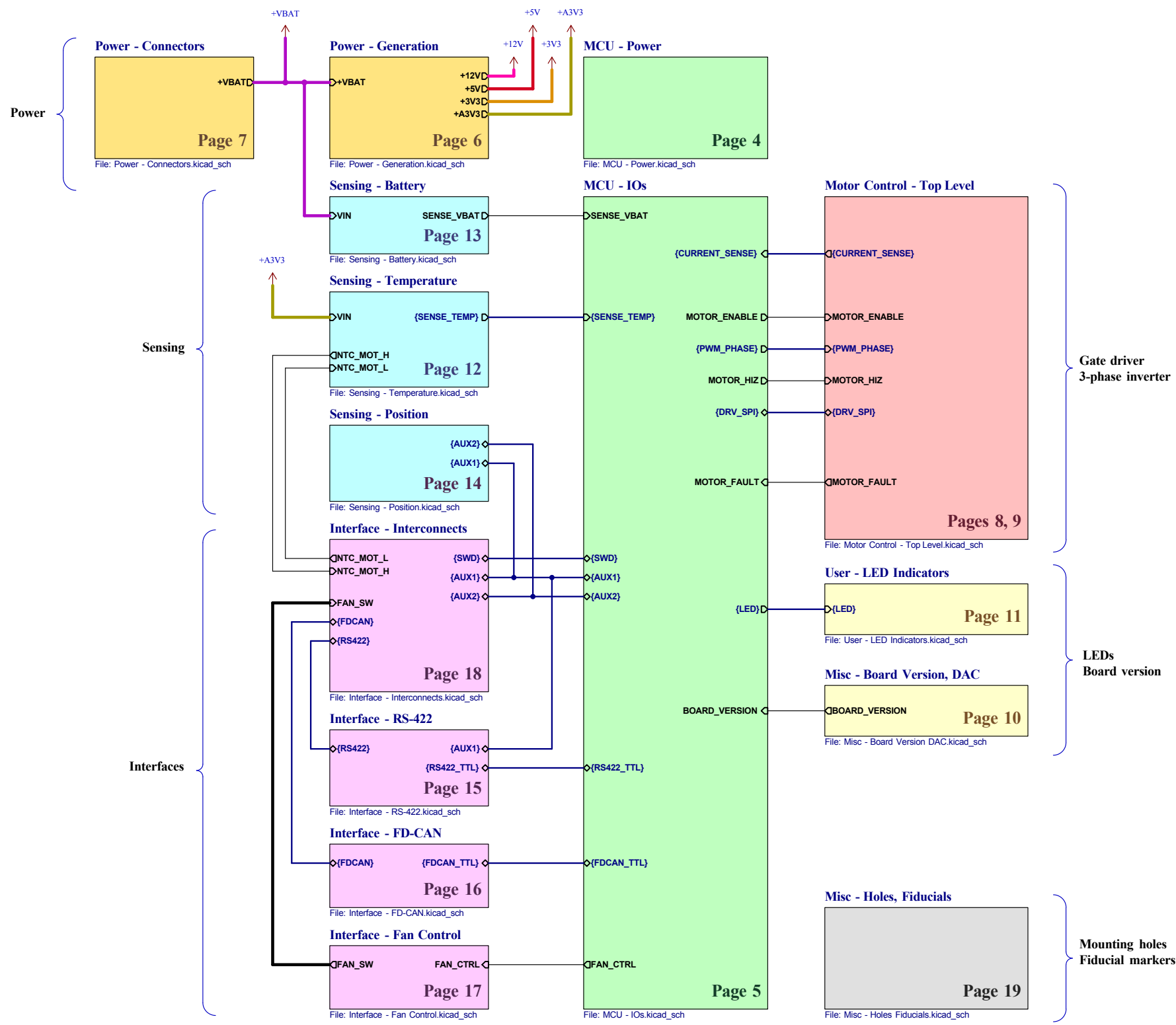
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	Sheet Path: /	File Name: amulet_controller.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13
			Revision: 1.1.1+ (Unreleased)	
			Size: A3	Sheet: 1 of 21

[2] Block Diagram

Page 4 (Power)
Page 5 (I/Os)

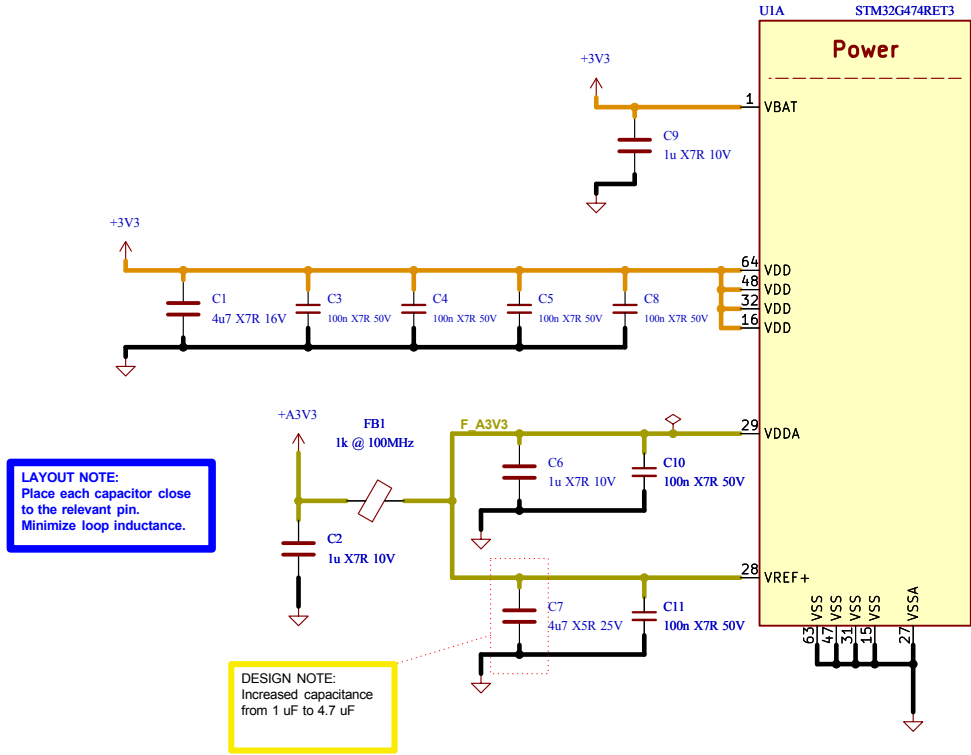


[3] Project Architecture



Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-22	Revision: 1.1.1+ (Unreleased)
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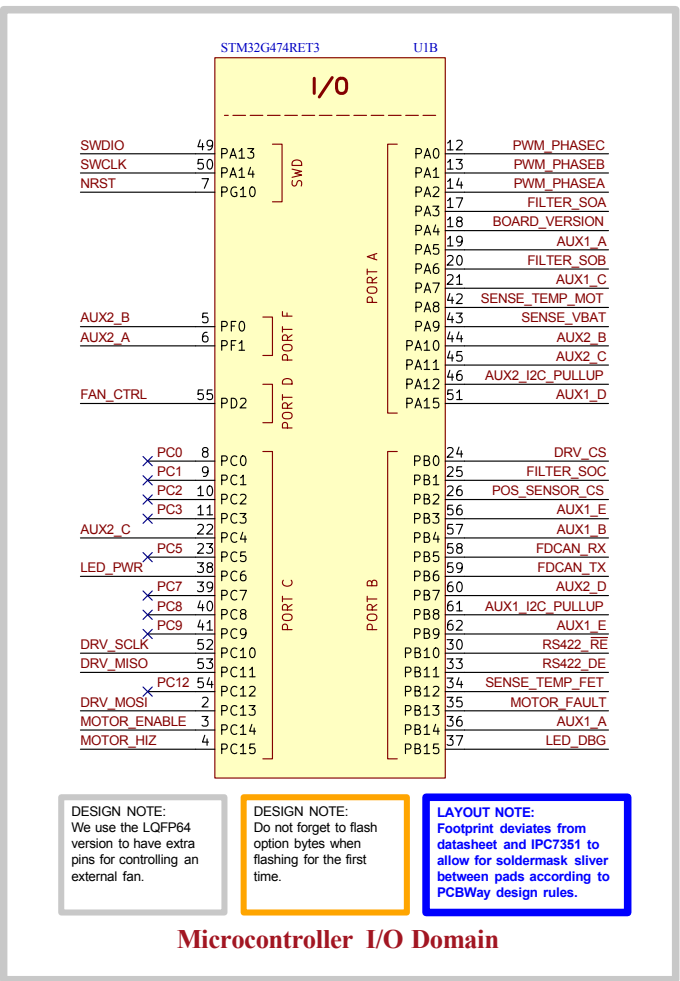
[4] MCU - Power



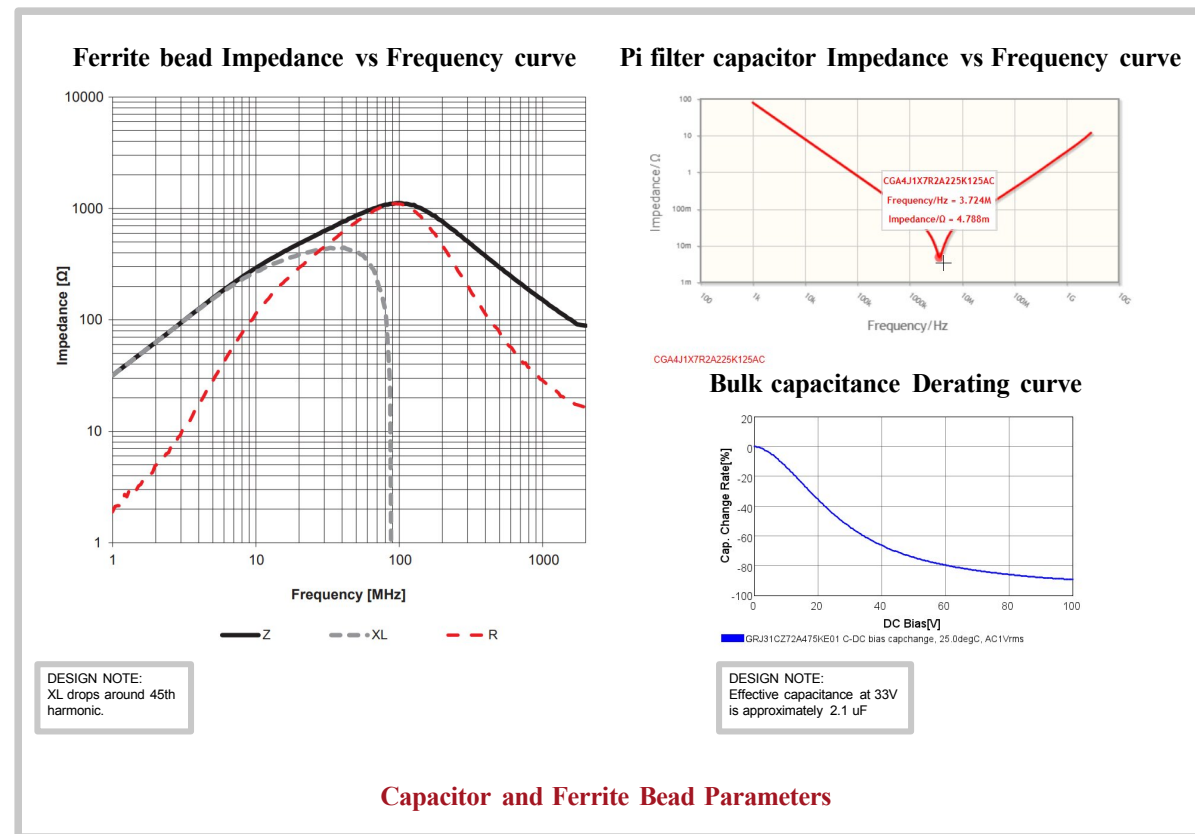
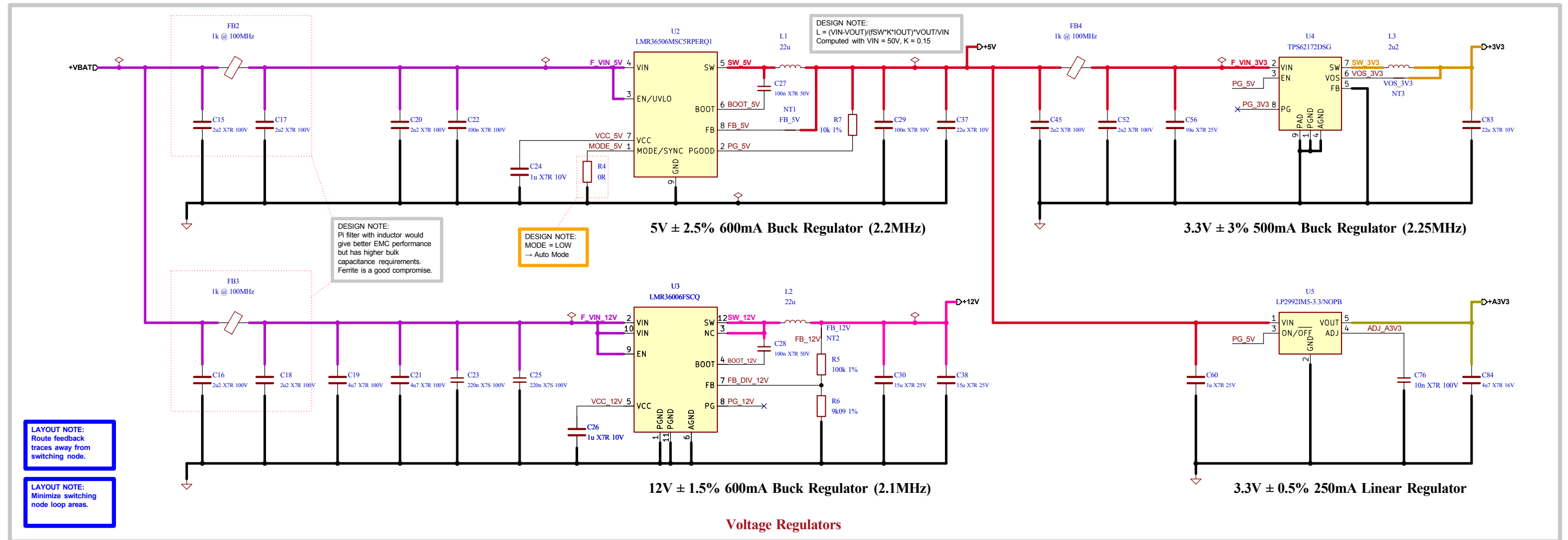
Microcontroller Power Supply Scheme

	Comments: AN5346 STM32G474 Datasheet p.81 J. Pieper ADC investigation	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
	Sheet Title: MCU - Power	Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Path: /Project Architecture/MCU - Power/	File Name: MCU - Power.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-18	Revision: 1.1.1+ (Unreleased)
			Reviewer:	Size: A4	Sheet: 4 of 21

[5] MCU - I/Os



[6] Power - Generation



Battery

- +VBAT TP39 +VBAT
- +VBAT TP40 +VBAT
- +VBAT TP41 +VBAT

Filtered Power Rails

- F_VIN_12V TP42 F_VIN_12V
- F_VIN_5V TP43 F_VIN_5V
- F_VIN_3V3 TP44 F_VIN_3V3

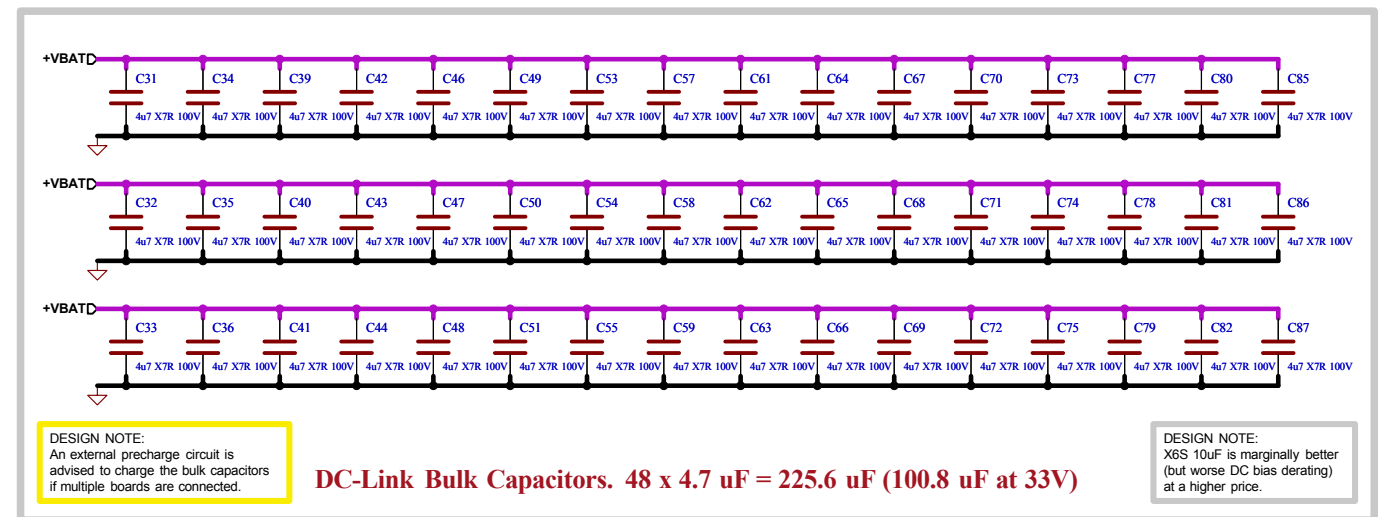
Power Rails

- +12V TP45 +12V
- +5V TP46 +5V
- +5V TP47 +5V
- +3V3 TP48 +3V3
- +3V3 TP49 +3V3
- +3V3 TP50 +3V3
- +A3V3 TP51 +A3V3

Ground

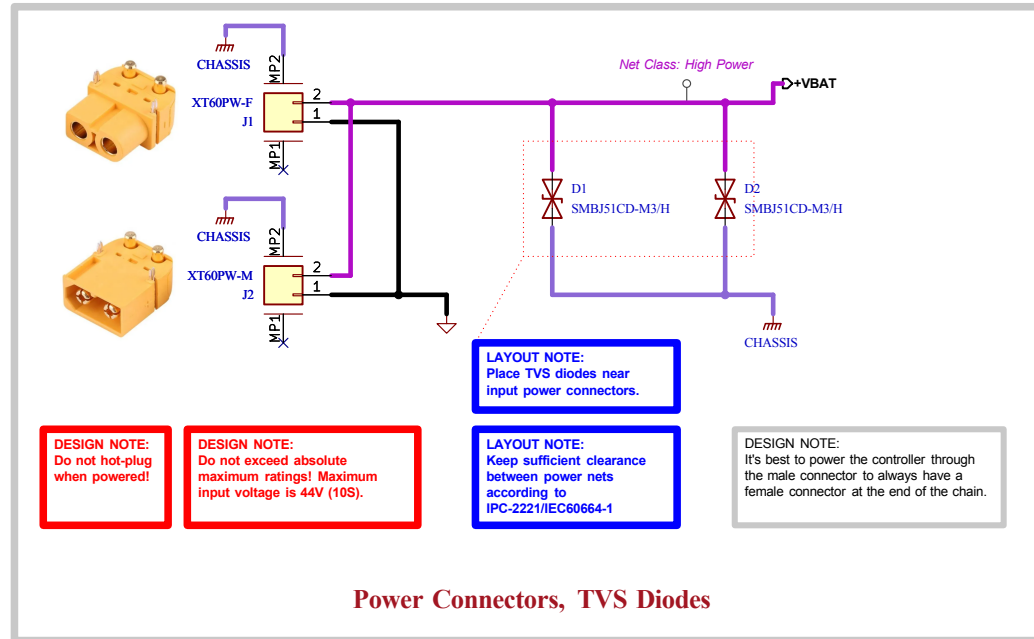
- GND TP52 GND
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- GND TP58 GND

Test Points



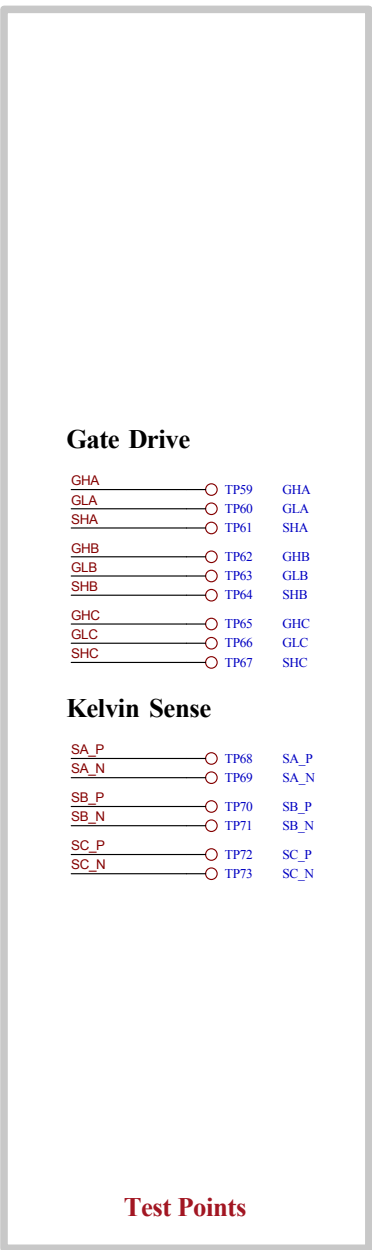
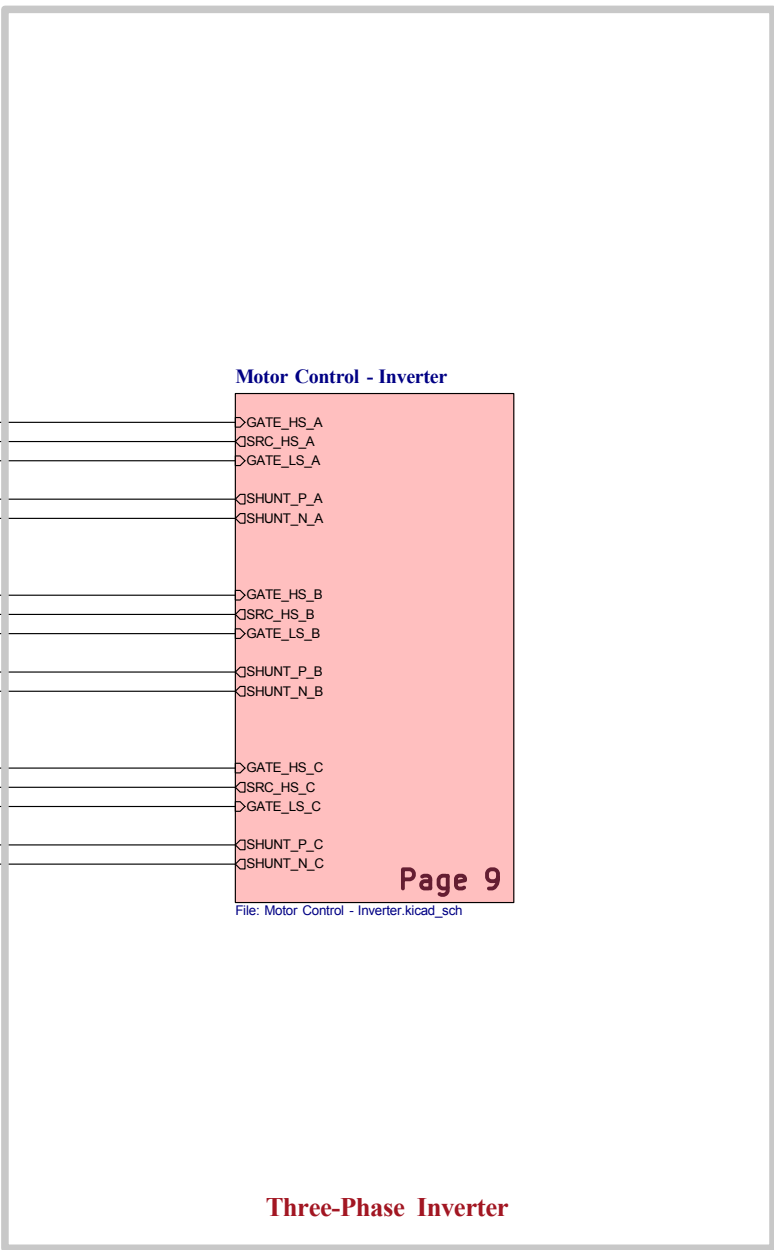
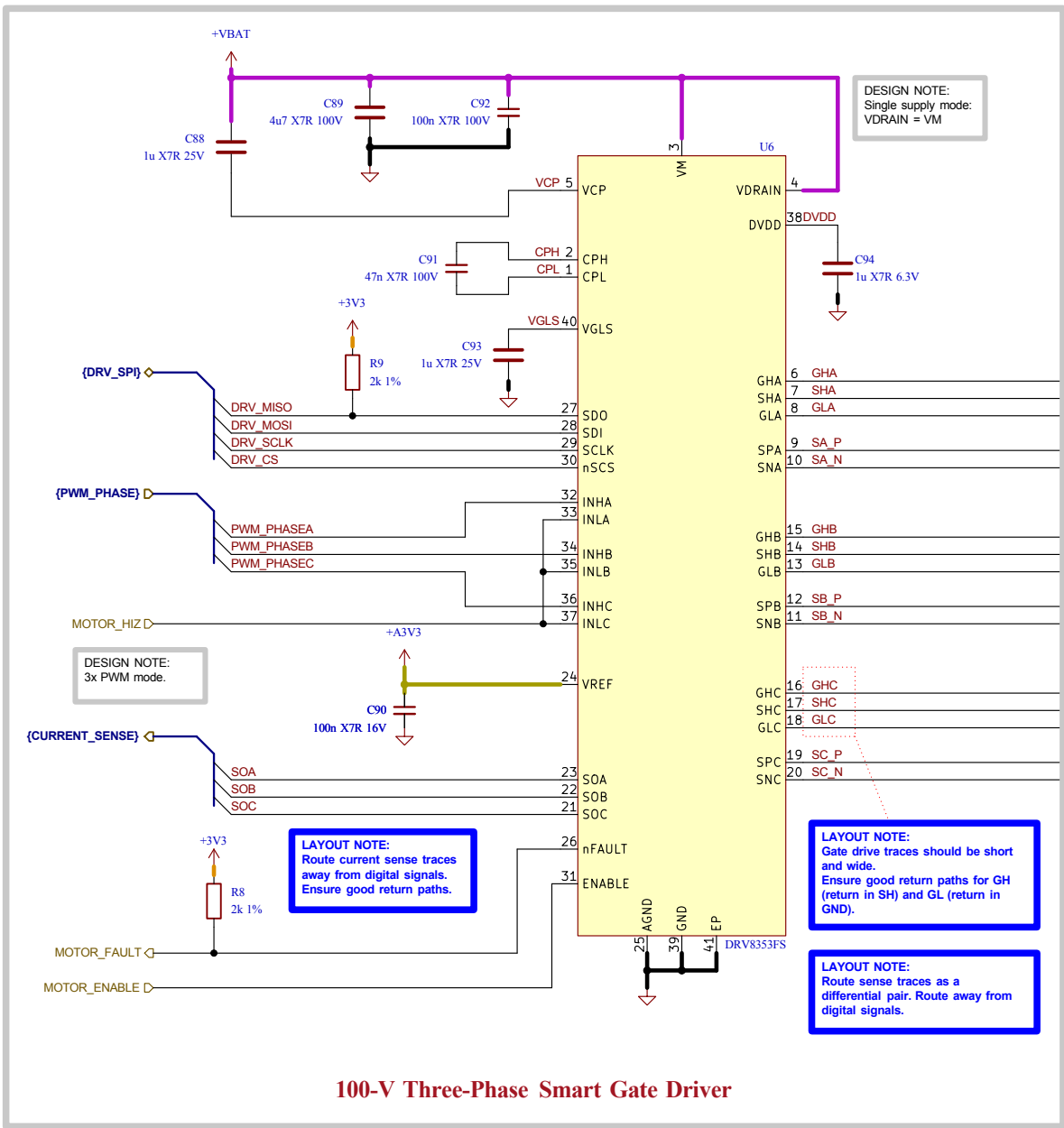
Comments: +12V (600 mA): Fan power +5V (600 mA): Gate Driver, LEDs +3V3 (500 mA): digital 3.3V +3V3A (250 mA): analog 3.3V		Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
Sheet Title: Power - Generation		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
Sheet Path: /Project Architecture/Power - Generation/		File Name: Power - Generation.kicad_sch	Designer: Vincent Nguyen	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
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[7] Power - Connectors



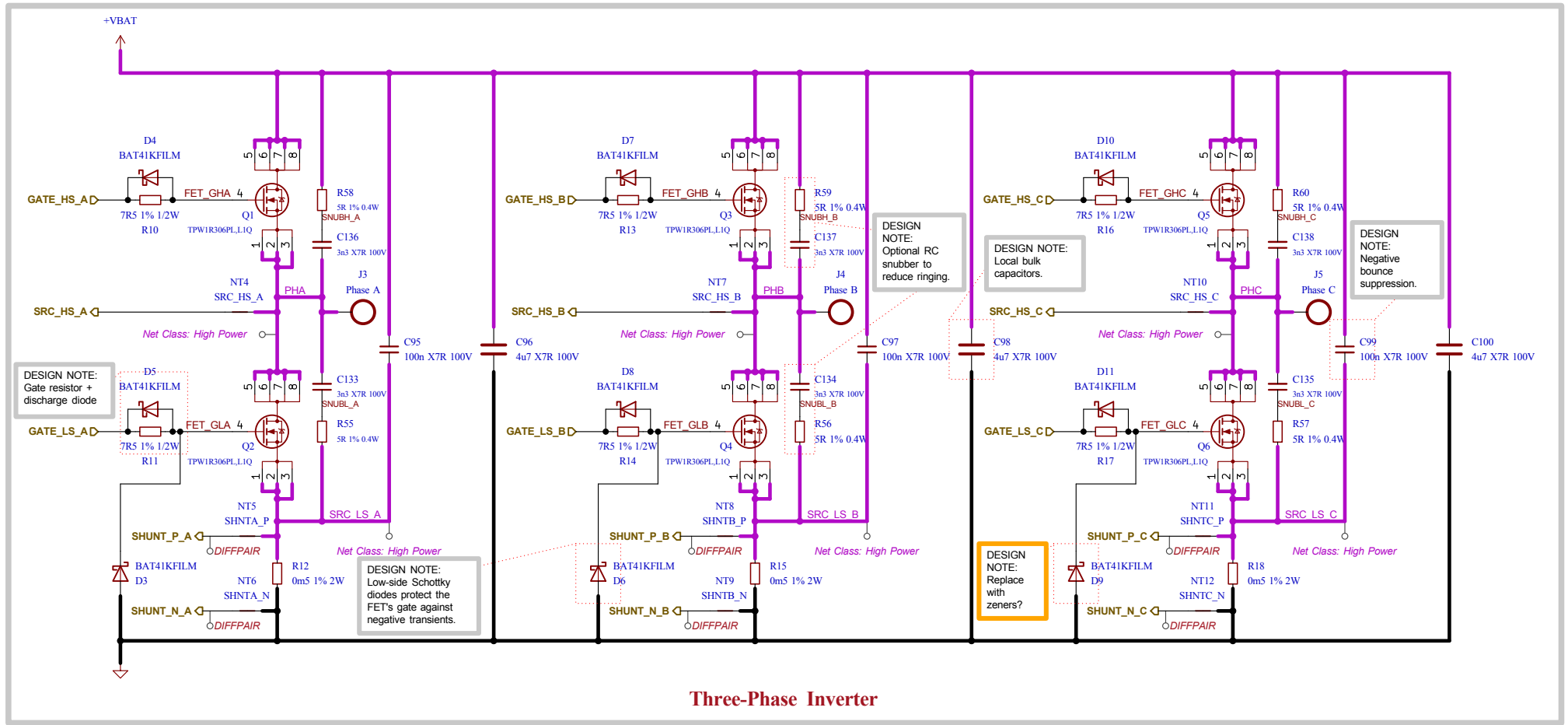
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[8] Motor Control - Top Level



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	Sheet Title: Motor Control - Top Level	Board Name: Amulet Motion Controller	Project Name: Chienpanzé	
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			Revision: 1.1.1+ (Unreleased)	
			Size: A3	Sheet: 8 of 21

[9] Motor Control - Inverter



LAYOUT NOTE:
High current traces must be carefully designed. Ensure ground return path does not cross sensitive parts of the board. Use multiple planes for higher current carrying capacity.

LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1.

DESIGN NOTE:
A gate drive current that is too large can damage the FETs!

Comments:
System Design Considerations for High-Power Motor Driver Applications
Best Practices for Board Layout of Motor Drivers
Proper RC Snubber Design for Motor Drivers

Sheet Title:
Motor Control - Inverter

Sheet Path:
/Project Architecture/Motor Control - Top Level/Motor Control - Inverter/

Company:
EPFL Xplore Research

Board Name:
Amulet Motion Controller

File Name:
Motor Control - Inverter.kicad_sch

Designer:
Vincent Nguyen

Reviewer:

Variant:

Git Hash:
5b68472

Project Name:
Chienpanzé

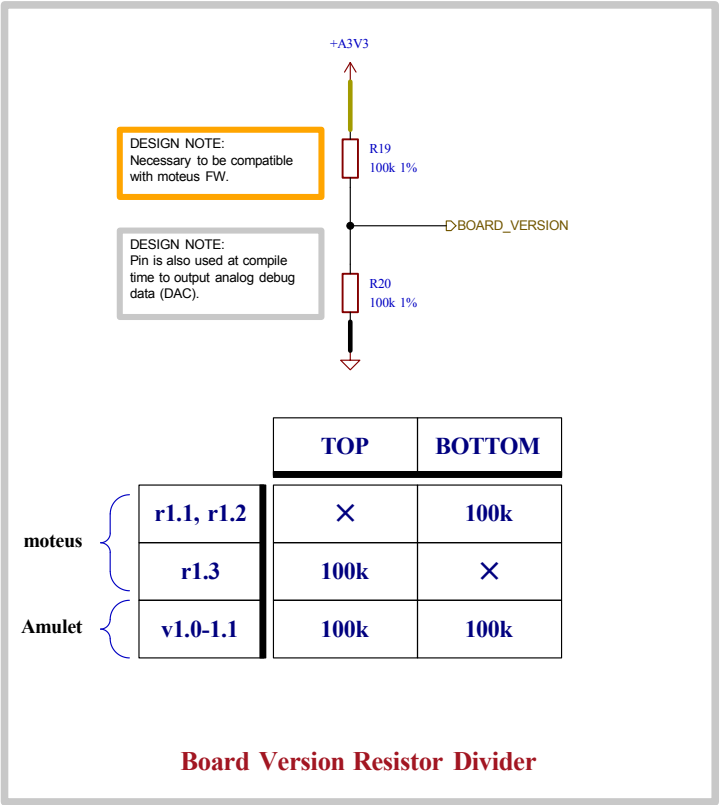
Date:
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Revision:
1.1.1+ (Unreleased)

Size:
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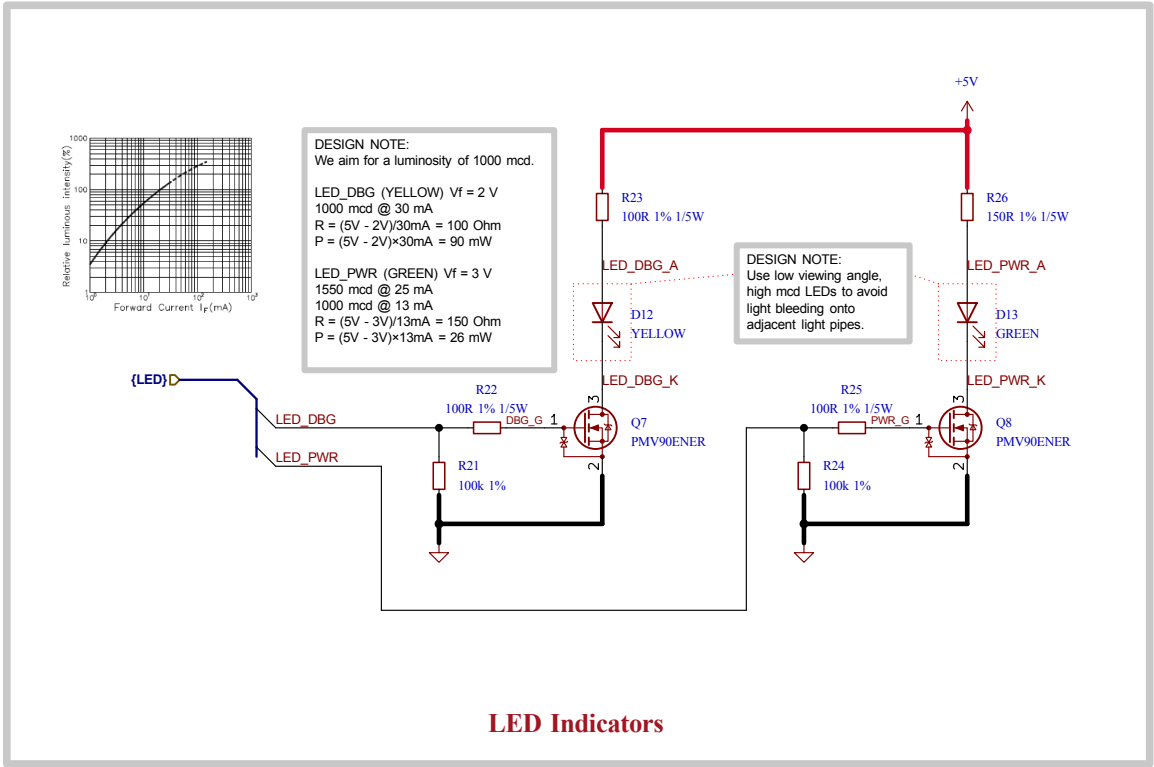
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9 of 21

[10] Misc - Board Version, DAC



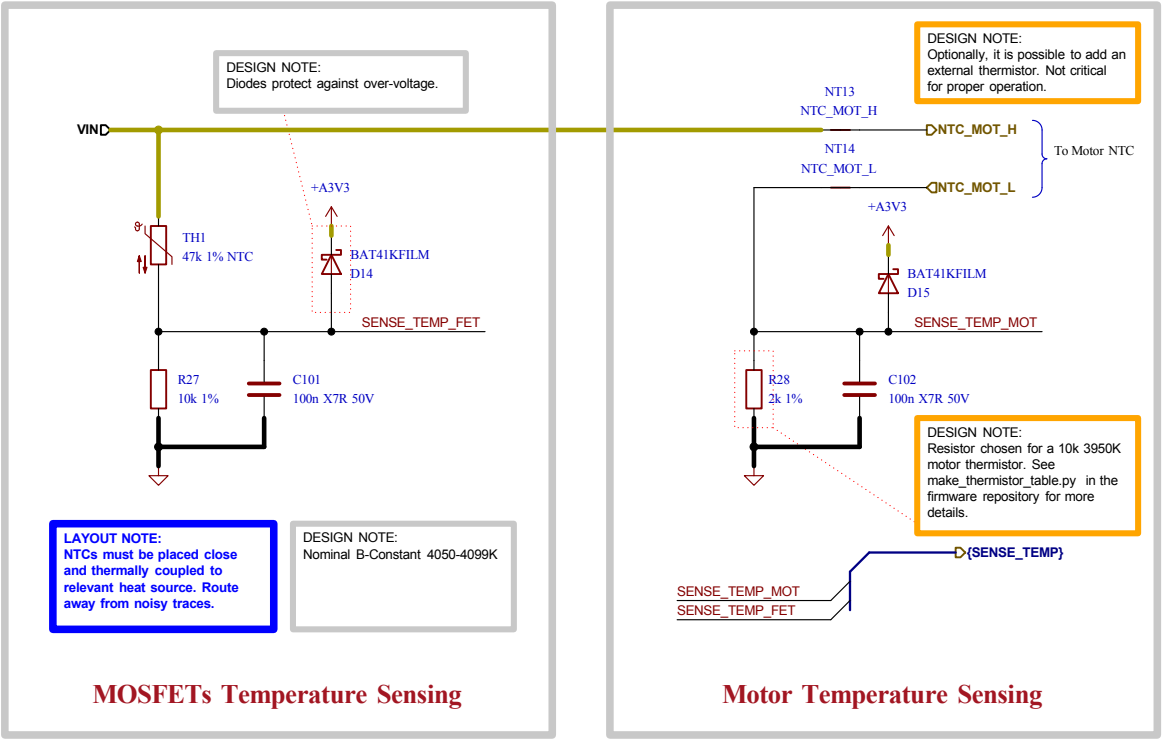
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[11] User - LED Indicators



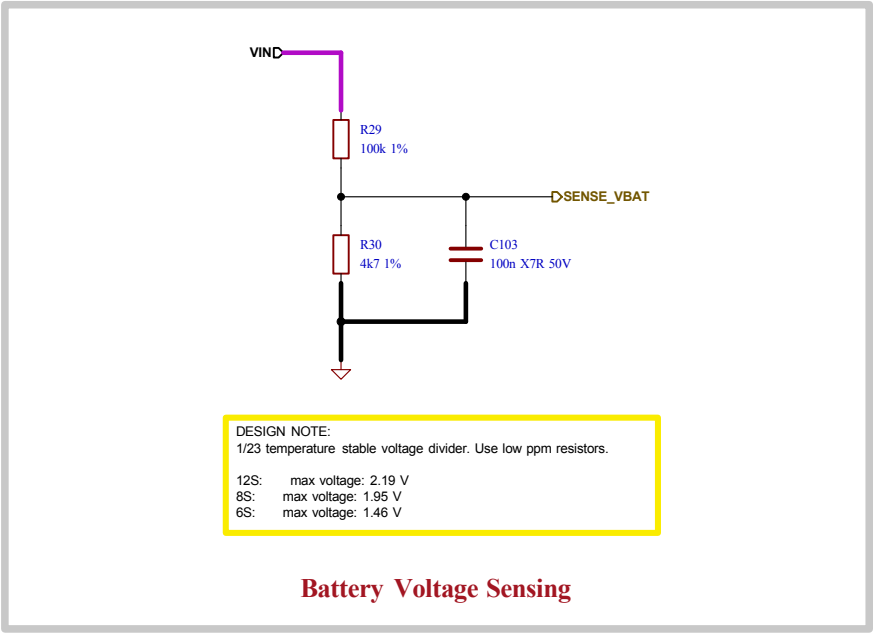
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[12] Sensing - Temperature



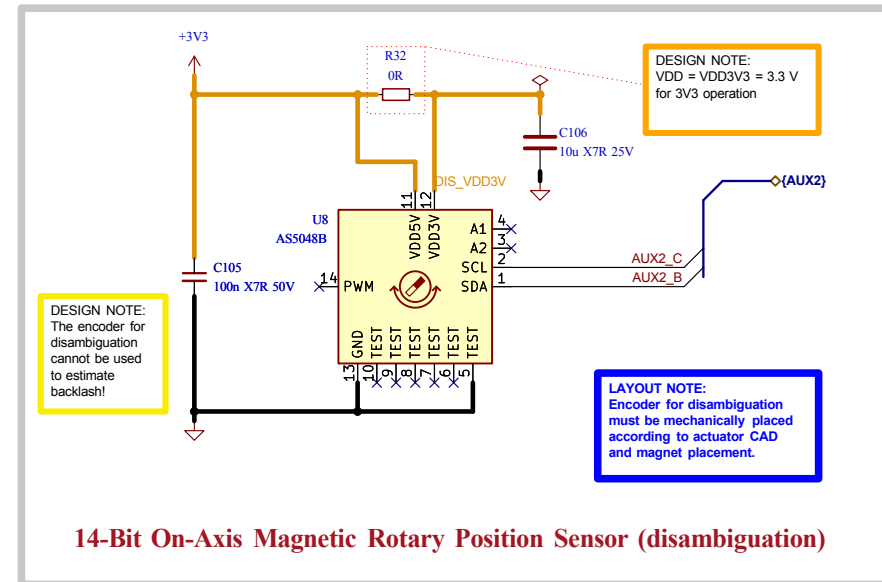
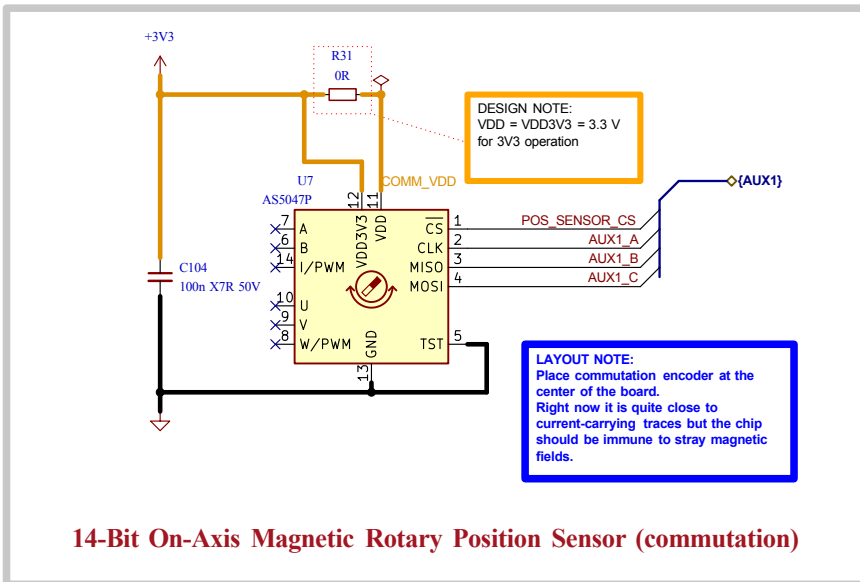
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				Sheet: 12 of 21

[13] Sensing - Battery



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		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
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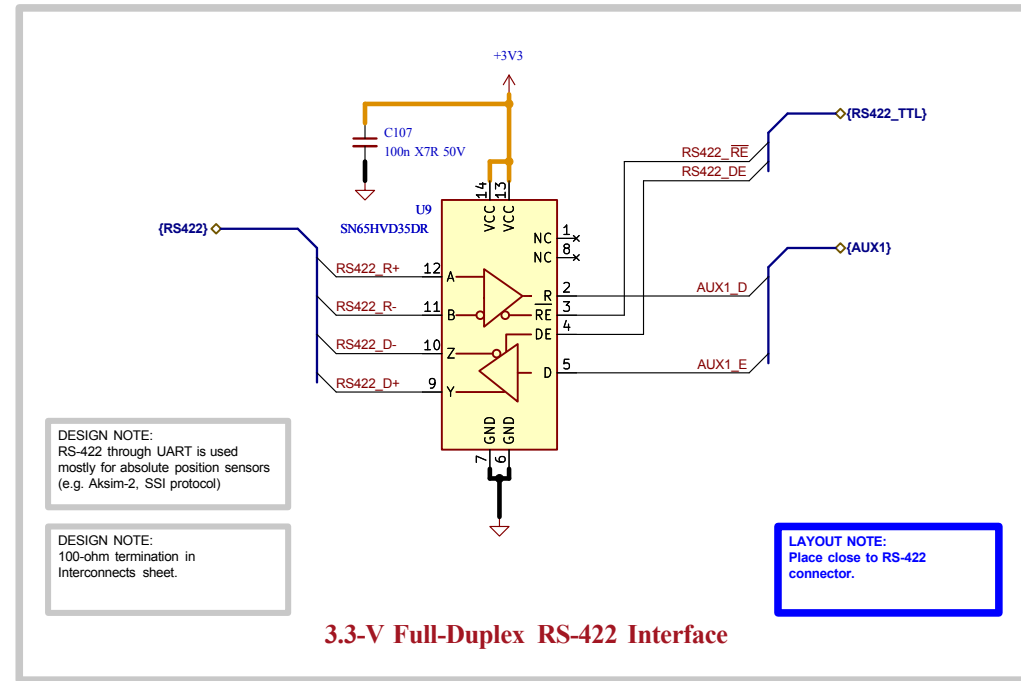
[14] Sensing - Position



DESIGN NOTE:
AS5047P senses magnet mounted on planetary sun gear, for commutation.
AS5048B senses magnet mounted on shaft with same reduction factor as planetary gearbox for disambiguation.

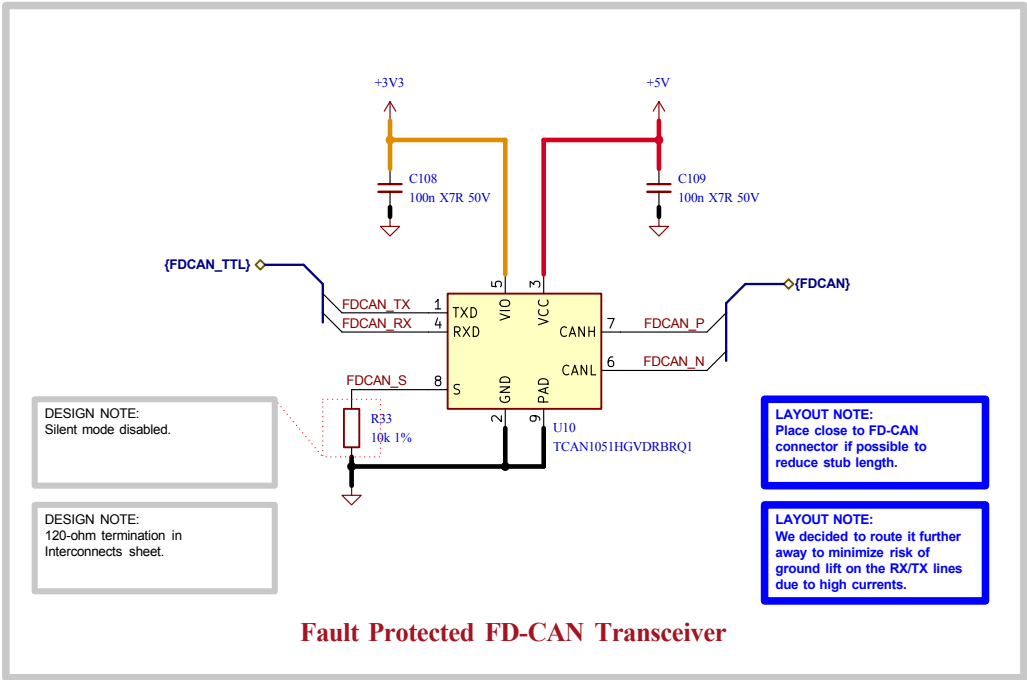
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			Reviewer:	Revision: 1.1.1+ (Unreleased)
			Size: A4	Sheet: 14 of 21

[15] Interface - RS-422



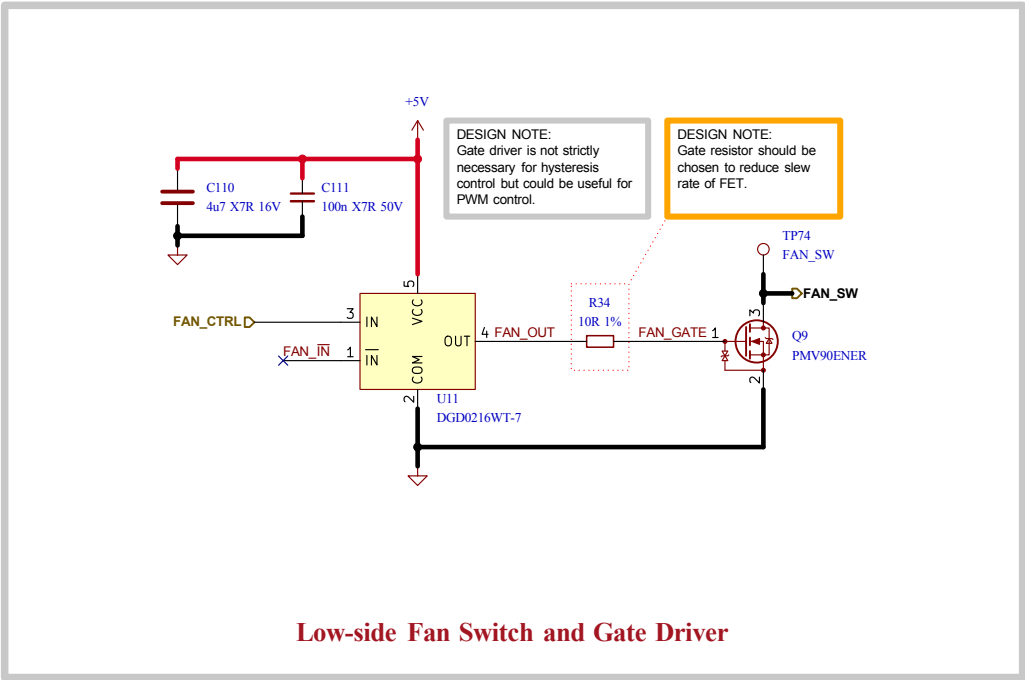
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[16] Interface - FD-CAN



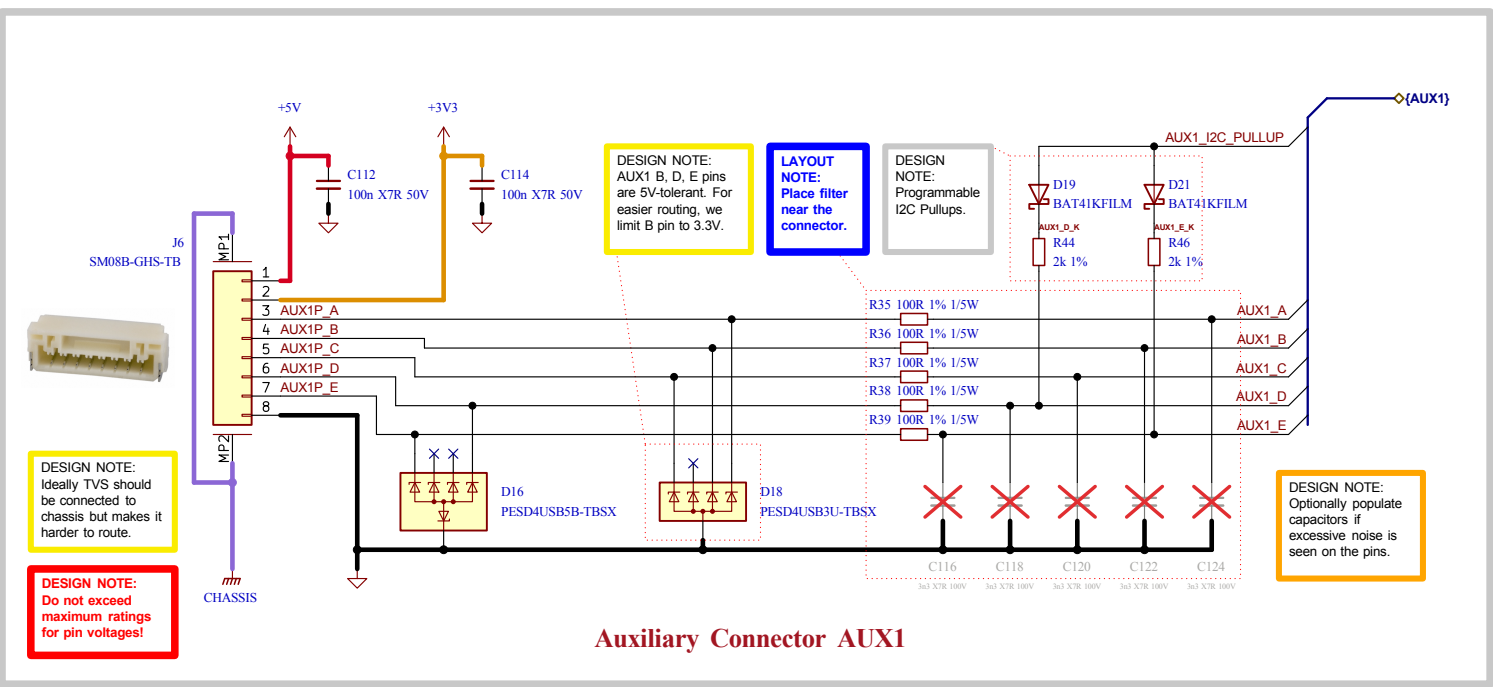
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[17] Interface - Fan Control

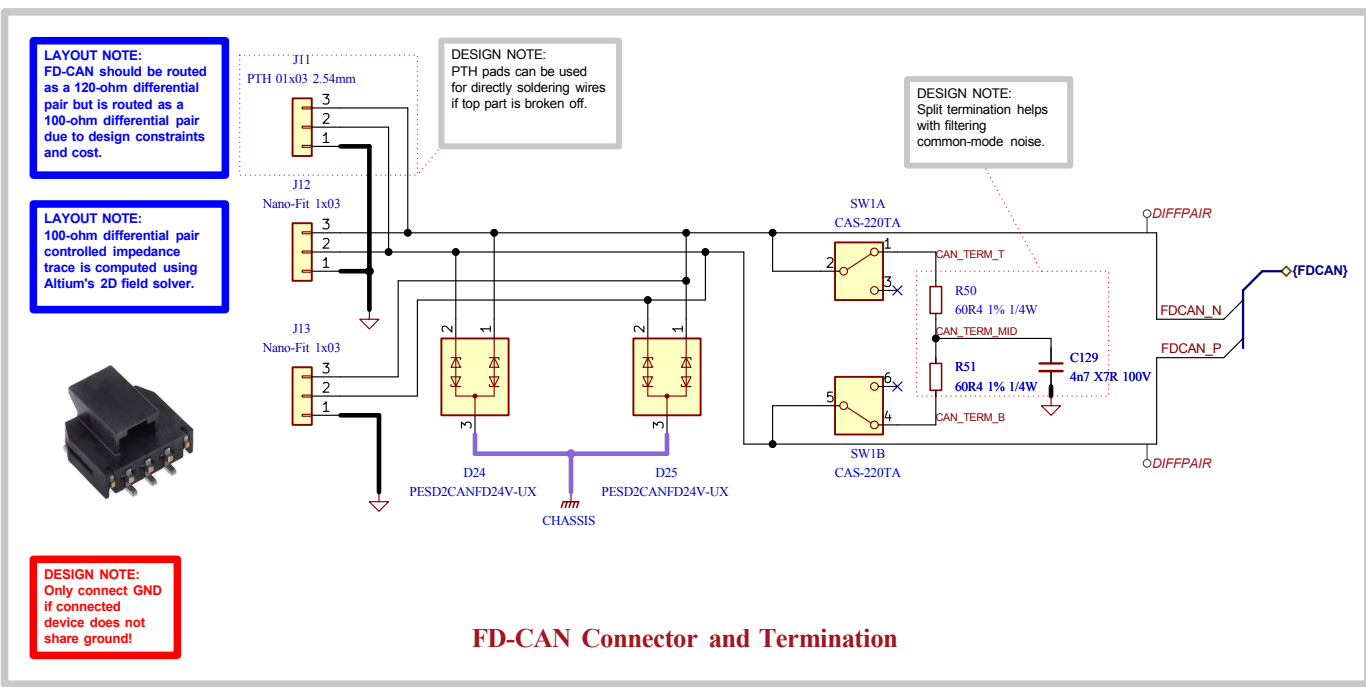


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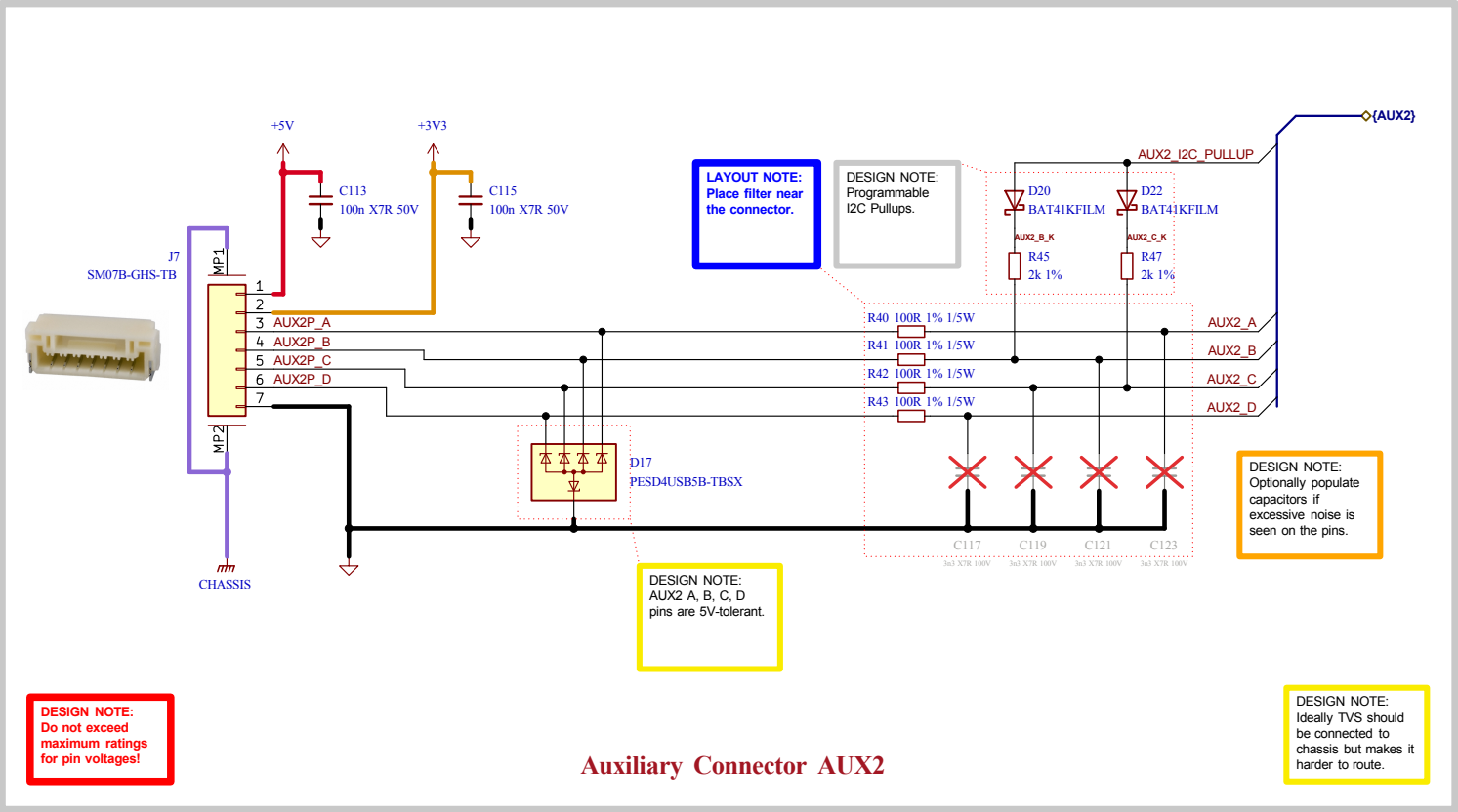
[18] Interface - Interconnects



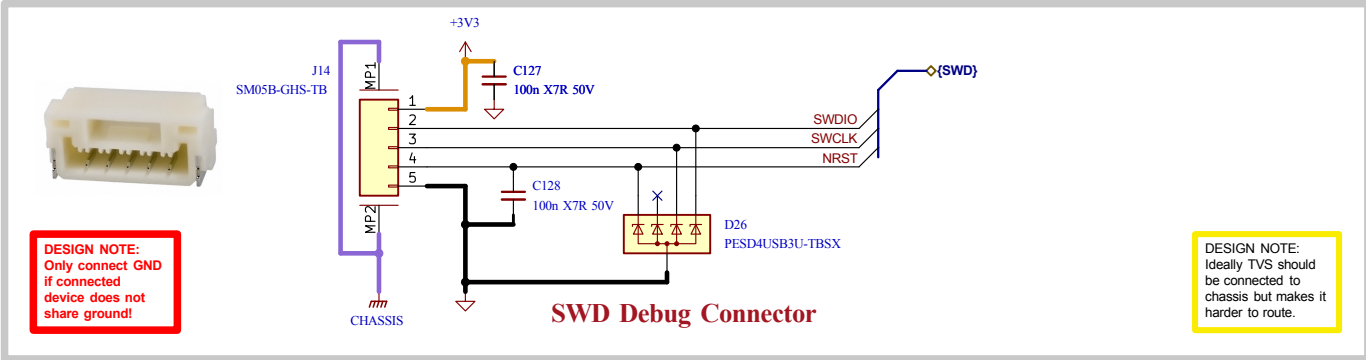
Auxiliary Connector AUX1



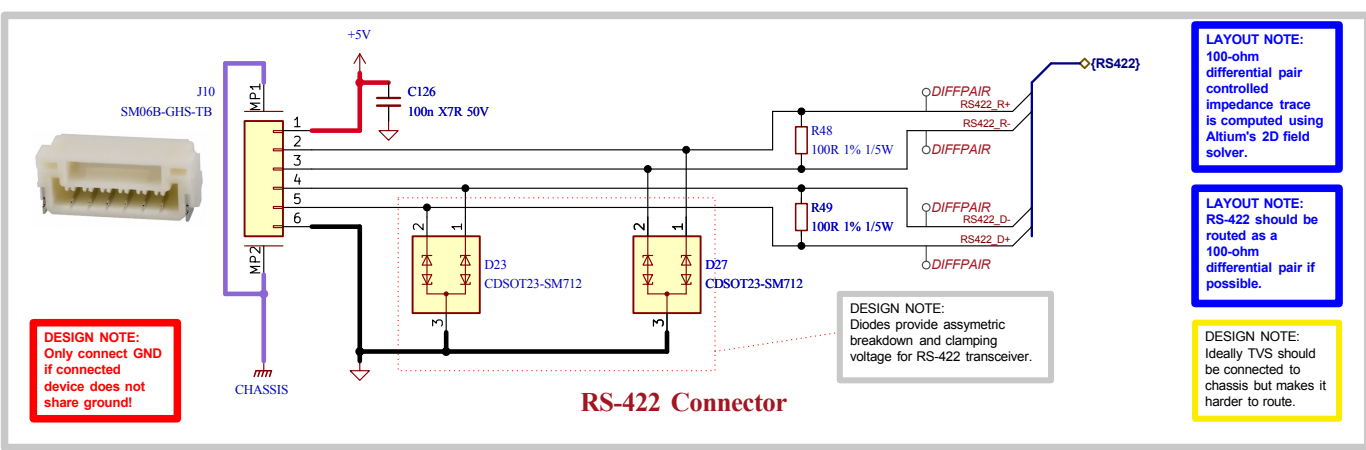
FD-CAN Connector and Termination



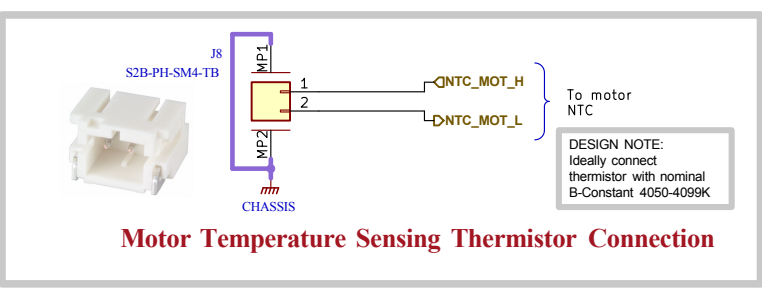
Auxiliary Connector AUX2



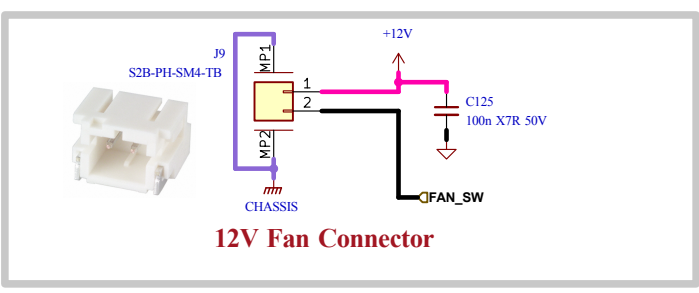
SWD Debug Connector



RS-422 Connector



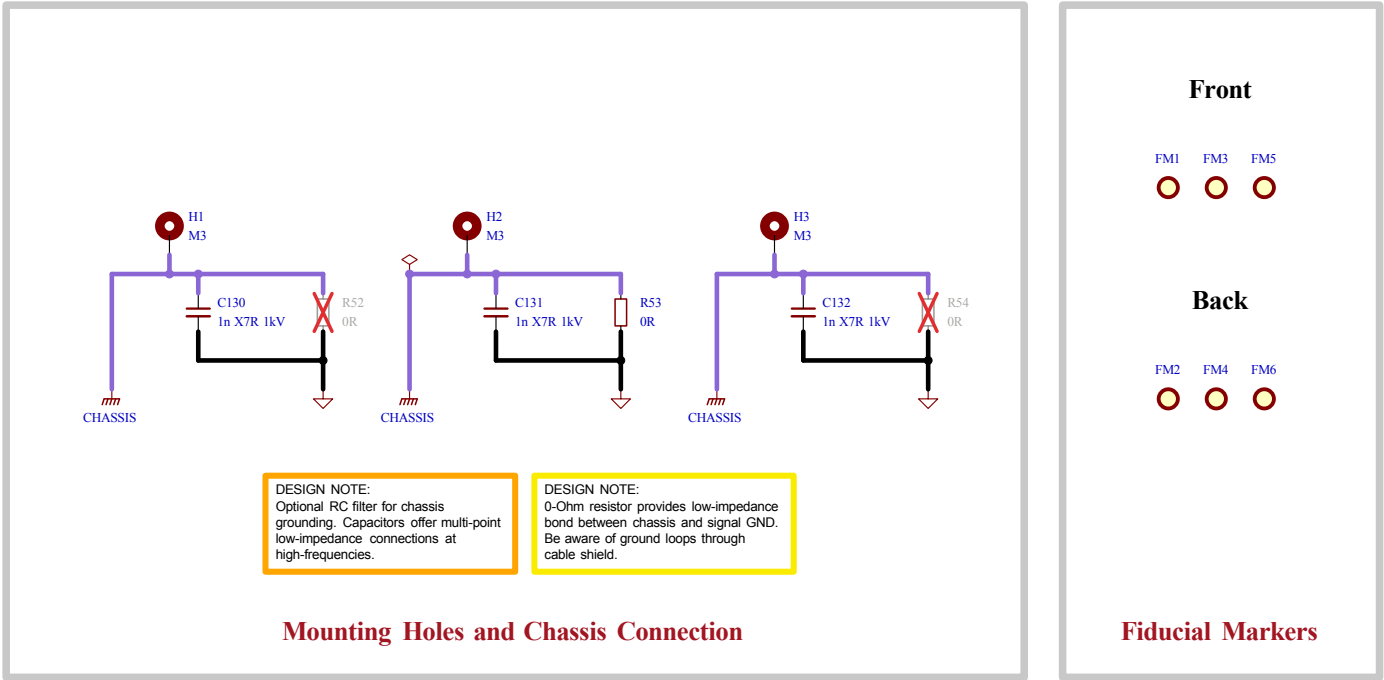
Motor Temperature Sensing Thermistor Connection



12V Fan Connector

	Comments: Reference: Flexible I/O worked examples	Company: EPFL Xplore Research	Variant:		Git Hash: 5b68472
		Board Name: Amulet Motion Controller	Project Name: Chienpanzé		
	Sheet Title: Interface - Interconnects	File Name: Interface - Interconnects.kicad_sch	Designer: Vincent Nguyen	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
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[19] Misc - Holes, Fiducials



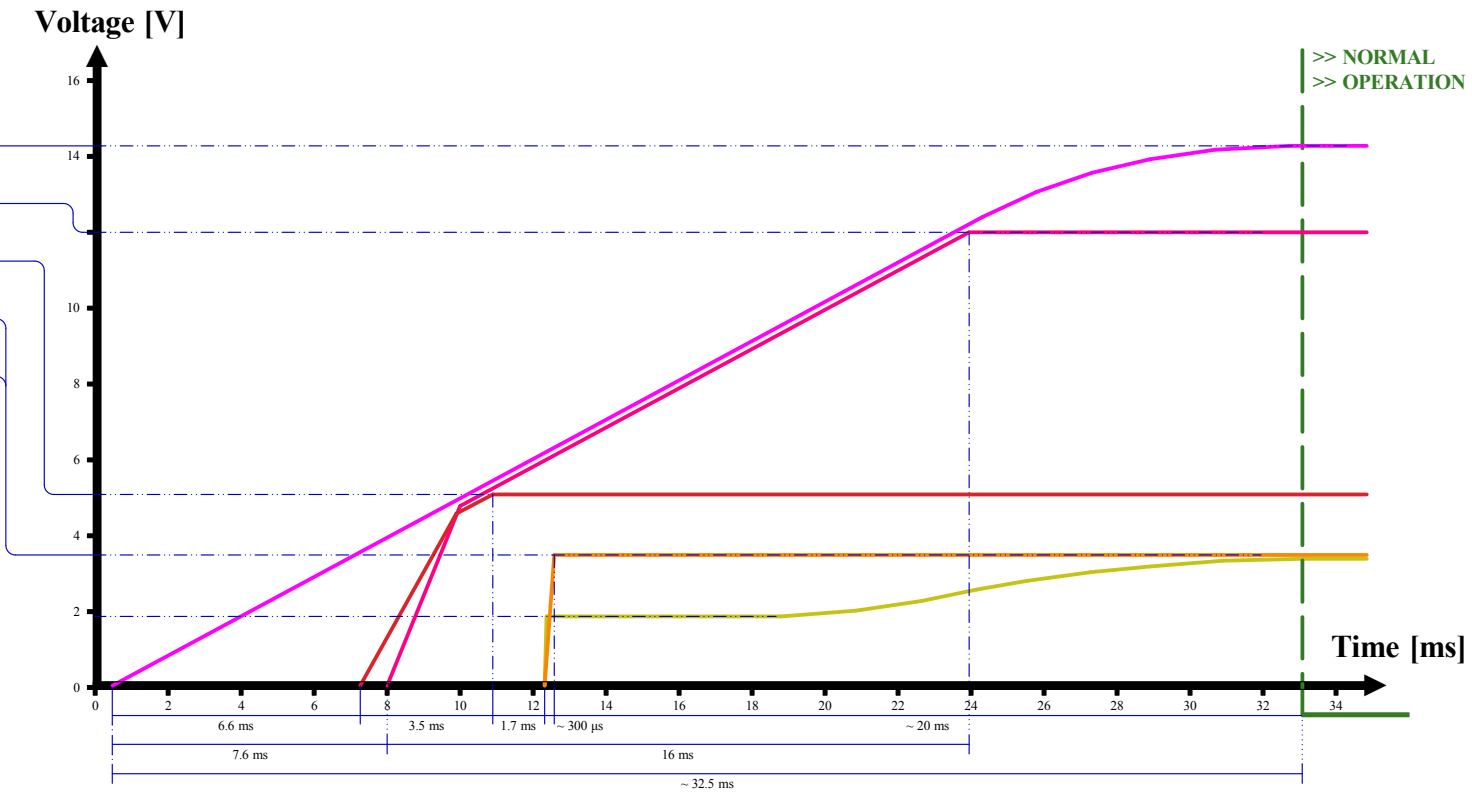
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	Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 21

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V ± 1.5%
+5V	LMR36506	5V ± 1.5%
+3V3	TPS62172	3.3V ± 3%
+A3V3	LP2992	3.3V ± 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
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	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 21

[21] Revision History

Version 1.0.0 - 2023-12-12

Added

- TVS protection and termination switch to FD-CAN.
- Low-side switched 12V 600 mA source for external fan.
- LDO for analog supply.
- One TVS diode per power connector.
- Second on-board I2C magnetic encoder for disambiguation.
- ESD protection to all interfaces.
- Over-voltage protection on thermistor ADC inputs.
- Pi filters to inputs of buck regulators and MCU analog supply.
- Decoupling caps next to power pins of connectors.

Changed

- CPH-CPL capacitor to 47 nF (gate driver).
- FD-CAN transceiver IC.
- FETs for top cooled variant.
- Input power TVS diode to bidirectional.
- Moved SOx low-pass filter to MCU section.
- PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- RS422 pinout on connector.
- Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

Added

- Controller target specifications.
- Credits to moteus on cover page.
- Optional RC-Snubber to power stage.

Fixed

- Chassis guard ring to go around the board.
- CAN and power TVS diodes now go to chassis.
- Clearance between nets to respect IEC60664-1 where possible.
- Comment on precharge.
- Power TVS diode reference designator from "U" to "D".

Changed

- 5V 300 mA buck converter with 600 mA version.
- Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

Changed

- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

Added

- RC snubber passive values.

Fixed

- More vias for VBUS and LMR36006 GND pads.
- Board version voltage reference from +3V3 to +A3V3.

Changed

- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

Fixed

- Replace non-stocked components.

	Comments:	Company: EPFL Xplore Research		Variant:	Git Hash: 5b68472
		Board Name: Amulet Motion Controller		Project Name: Chienpanzé	
	Sheet Title: Revision History	File Name: Revision_History.kicad_sch	Designer: Vincent Nguyen	Date: 2024-01-03	Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 21